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A Formula for Investing in the Renewable Energy Sector

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Abstract

The renewable energy sector has seen substantial growth in recent years, presenting not only a critical shift towards sustainable development but also encouraging individuals to invest in accordance with ethical and environmental considerations with the potential for financial returns. Along with these opportunities comes the challenges in selecting high-yield stocks, a task which for many can be both intimidating and unrewarding as it requires knowledge, time, and dedication. While there are thousands of investment strategies, few are publicly available in the renewable energy sector, thus leaving a gap in research. The purpose of this paper is to develop a method which simplifies the process of selecting high yield renewable energy stocks on the global stock market. Drawing inspiration from Joel Greenblatt's magic formula and implementing panel data regression based on 10 stocks from 2007-2023, return on asset and revenue growth were shown to be statistically significant. A portfolio of 6 stocks was created based on a subset of 35 stocks in the renewable energy sector from 2018-03-27 to 2023-03-27. At an annual basis the portfolio was rebalanced based on highest return on asset and revenue growth. The portfolio outperformed the market index for global clean energy and the top-performing clean energy exchange-traded fund but did not generate any excess return.

Key words: Magic Formula, Renewable energy sector, Value Investing, Quality investing, Efficient Market Hypothesis, CAPM.

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1 Introduction

1.1 Background

In recent years, the renewable energy sector has witnessed substantial growth and attention due to its pivotal role in the development of our society and the global economy. The sector increased its energy capacity by 50% from 2022 to 2023, and it is forecasted to reach a growth factor of 2.5 by 2030 (*Fastest growth rate for renewable energy in two decades*, 2024). At the same time the total return of Nasdaq and S&P 500 was 44.7% and 26.3% respectively (*2023 Review & 2024 Outlook | Nasdaq*, no date). This significant growth in the renewable energy sector presents not only a critical shift towards sustainable development but also offers opportunities for individual investors. Investing in the sector not only aligns with ethical and environmental considerations but also has the potential for financial returns.

While there are thousands of investment strategies, few are publicly available in the renewable energy sector. Published research evaluates returns on renewable energy firms (Inchauspe, Ripple and Trück, 2015) and investment strategies using factor models that utilizes Exchange-Traded Funds that track clean and renewable energy firms (Miralles-Quirós and Miralles-Quirós, 2024). There is a research gap concerning investment strategies that use traditional financial metrics to value such firms while simultaneously being easily comprehensible by the average investor. In his book “The little book that still beats the market”, Joel Greenblatt motivates his creation of the magic formula as a simple yet effective tool for novice investors to enter and outperform the stock market. However, his formula does not include utilities stocks thus excluding renewable energy firms (Greenblatt, 2010, p.xvii, 142). This creates a barrier of entry for individual investors as they cannot use his Magic Formula for investing in this specific sector. Instead, investors have to conduct their own research and estimate financial metrics that may be a good measure of price for these stocks, which is difficult to do.

Therefore, this paper seeks to create an investment method that is easy to understand, outperforms the top-performing exchange-traded fund, and that specifically focuses on clean energy investments. Drawing inspiration from Joel Greenblatt’s renowned Magic Formula, I will devise a methodology specifically for the renewable energy sector. Panel data regression is used to determine which financial metrics can explain positive movement in stock price, and a portfolio will be created based on the significant variables. The investment formula aims to beat the clean energy market and generate excess. Finally, the Capital Asset Pricing Model

(CAPM) will be used to determine if the portfolio generates excess return and in turn if it disproves the Efficient Market Hypothesis.

1.2 Research Question

1. Which financial metrics should be implemented into a formula to select the top-performing renewable energy stocks?
2. Does the Investment Formula outperform the top-performing renewable energy Exchange-Traded fund?
3. Does the Investment Formula generate excess return?

1.3 Delimitation

Due to a large number of renewable energy firms, the focus has been limited to stocks in the American, Canadian, French, German, Swedish and Norwegian stock market. Greenblatt suggested that the Magic Formula will only work on firms with a market cap of < \$50 million, which was the second limitation for the data collection.

Only quantitative methods will be used in deriving an investment formula as qualitative variables might introduce ambiguity and is not theoretically aligned with the derivation of the Magic Formula and its applications. For example, the Magic Formula contains two quantitative financial metrics (earnings yield and return on tangible capital), and then selects stocks based on the firms with the highest value of these metrics. A qualitative approach, on the other hand, would involve analysing a firm's business idea, marketing strategy and overall management effectiveness to determine its potential growth and profitability. While a qualitative approach may improve the ability to select high-yielding stocks, it would increase the difficulty for the average investor to do so if it would involve complex business analysis. Lastly, the time-limitation was from 2007-2023 as to capture the stock's performance in economic upswings and downturns.

2 Previous research

2.1 Magic Formula: Empirical data

The Magic formula is a value-based investment strategy created by Joel Greenblatt which aims to consistently beat the stock market by purchasing undervalued stocks at a discount. It includes two key financial metrics: earnings yield (EBIT/EV) and return on assets (ROA). According to this strategy, ROA should be used as a screening criterion set at the minimum of 25%. Greenblatt does not provide a specific percentage for earnings yield. (Greenblatt, 2010).

In 2018, a study from Lund university backtested the Magic Formula against the OMX on the Swedish stock market. The results showed an annual return of 5.22% for the market while the Magic Formula had an annual average of 21.25%. The data was based on returns from 2007-2017 (Gustavsson and Strömberg, 2018). Similar results have also been found on the Finnish stock market two years prior where the Magic Formula beat the market and yielded abnormal returns which were not compensated for higher level of risk (Davydov, Tikkanen and Aijo, 2016).

Furthermore, the Magic Formula has also been backtested on the entire Nordic Region from 1998 to 2008. These results show, with a risk-adjusted portfolio with CAPM and Fama-French's three factor model, that unlike previous studies, the formula did not yield any excess return. It did however still perform better than the market. The same study also used the Magic Formula to conduct a backtest on the Norwegian Stock Market from 2003 to 2022. There they did observe excess return. However, the inclusion of transaction costs reduced the significance level from $p < 0,05$ to $p < 0.10$ which suggest that these results are not achievable in real-worlds conditions (Vestre and Wikheim, 2022).

These studies suggest that the Magic Formula outperforms the market overall but with some limitations including transaction costs. To address this, I will select six stocks, thereby reducing the amount of transaction costs one is exposed to.

Due to the varied results of these backtests, some scholars are critical of the Magic Formula. For example, Victor Persson and Niklas Selander recommended that the investment strategy is better used as a screening tool. They emphasize that further research is needed before investing based solely on the Magic Formula, suggesting caution (Persson and Selander, 2009). Another approach is to modify the Magic Formula and improve results. This has been done previously by Douglas W Blackburn and Nusret Cakici, where they substituted EBIT/EV for gross profit. By doing this, they yielded significantly more favourable results. Furthermore, they observed

consistent and strong outperformance of profitable value stocks compared to unprofitable growth stocks across all of Asia, Europe, North America and Japan (Blackburn and Cakici, 2017).

With this research in mind, I will aim to either create a modified magic formula that includes additional financial metrics or create an entirely new investment formula. The decision depends on the results from the panel data regression. If my portfolio outperforms the market, in realized returns, I will suggest investors to use my formula without further research. If it does not, additional research is needed before investing.

2.2 Renewable energy sector: Performance indicators

While renewable energy stocks have garnered significant attention in recent years, there is a notable gap in existing research exploring the relationship between changes in stock price and financial metrics. However, previous research in other areas that are complementary to this paper have been researched and evaluated. A study published by Multidisciplinary Digital Publishing Institute showed that both direct and non-innovative subsidies have a significant effect on renewable energy firms' financial performance (Zhang *et al.*, 2015). Subsidies can indirectly increase the values of financial metrics that include revenue in the equation which in turn may affect the results of the investment formula developed in this paper. Subsidies also have a major effect on Research & Development (R&D), which is a key component for these firms as it is mainly technology that provide revenue (Sung, Choi and Song, 2019). Other factors, such as shareholder protection, show a positive significant effect on R&D (Karlilar and Tarzibashi, 2023).

The performance of renewable energy firms and their projects can both be contributed to financial- and non-financial factors. These two are positively interlinked which means an improvement in one is often associated with improvements in the other. Policy makers' support of these projects have a positive financial and non-financial effect, and investors improve the overall financial situation of these firms (Maqbool, Deng and Ashfaq, 2020). All of this together will have a major effect on renewable energy firms, and in turn their stock performance.

All factors above will not be included in the model and the investment formula; however, it is important to highlight that they have both indirect and direct effects on the renewable energy firm's financial performance, and in turn the stock's performance. In conclusion, these factors

might explain movement in stock price that my regression model and investment formula cannot.

3 Theory

In developing an investment formula, it is essential to ground it in financial theory. The following section introduces key theories such as the Efficient Market Hypothesis (EMH) and the Capital Asset Pricing Model (CAPM) which will be used to evaluate whether my portfolio generates excess return. Additionally, theories such as Value Investing, Quality Investing and the Magic Formula will act as a foundation behind the variables included in the regression model and later, the investment formula.

3.1 The Efficient Market Hypothesis

The efficient market hypothesis (EMH), developed by Eugene Fama, states that security prices at any time “fully reflect” all available information. Thus, investors can never outperform the market through investment strategy or market timing. “Fully reflect” is defined in many ways, and one such definition is that the expected return on securities is determined based on their perceived level of risk. This is defined as the market being “efficient” (Fama, 1970).

Eugene Fama suggest that a market is efficient under any of these three conditions:

- i) There are no transaction costs in trading securities.
- ii) All available information is available to all market participants at no costs.
- iii) There is no disagreement on the implications of current information impacts both the present price and the expected future price distribution of each security.

Additionally, Fama identifies three different information subsets: *weak form*, *semi-strong form*, and *strong form*. The forms are divided based on how efficient the market is and requires different methods of study based on this evaluation (Fama, 1970).

3.2 Capital Asset Pricing Model

The capital asset pricing model (Sharpe, 1964), was based on a model of portfolio choice developed by Harry Markowitz in the year 1959. Markowitz’s model estimates the mean-variance-efficient portfolio and aims to minimize the variance of portfolio return given expected return, and maximize return given the minimum variance. CAPM turns this algebraic condition into a testable model that explains the relationship between expected return and risk (Fama and French, 2004).

$$E(R_i) = R_f + [E(R_M) - R_f]\beta_{iM}, i = 1, \dots, N.$$

Beta (β) is a measure of the asset’s risk or risk of the market portfolio, measured by the variance of its return. The denominator is a weighted average of the covariance risk of the assets in M.

The risk-free rate (R_f) is defined as the return of an asset with a beta equal to 0 and is uncorrelated with market return. The market return (R_M) is the rate of return of a market (Fama and French, 2004).

CAPM can be rewritten to explain the relationship between expected return and market beta in a time-series regression. The equation suggests that the expected value of a security or asset's excess return ($R_{it} - R_{ft}$) is entirely explained by its expected risk premium (β times the expected value of $R_{Mt} - R_{ft}$). The alpha (α) or Jensen's alpha is the intercept and should be zero for each asset. However, if the alpha is positive, it indicates that the return of that asset or security cannot only be explained by the amount of risk that one is exposed to (Fama and French, 2004).

$$R_{it} - R_{ft} = \alpha + \beta_{iM}(R_{Mt} - R_{ft}) + \varepsilon_{it}$$

3.3 Value investing

Value investing, developed by Ben Graham and David Dodd, focuses on identifying stocks that are undervalued relative to their intrinsic worth or fundamental value. The goal is to buy these undervalued stocks and hold them until their true value is recognized by the market, leading to price appreciation. Indeed, empirical data suggests that, over time, firms trading at lower multiples, more sustainable cashflows, with solid balance sheets, higher profitability, and lower volatility and beta will lead to higher stock returns (Lee, 2014). The Value investing theory contradicts the CAPM and the EMH. If the market was efficient, there would not be any arbitrage opportunities and therefore the asset would be unable to generate excess return as the only way to generate above market return is to take on more risk.

Ben Graham and David Dodd emphasizes that value investing is not only a quantitative method but also a qualitative one (Graham and Dodd, 2023). However, we will only base our method and findings on the quantitative aspect of value investing, using different financial metrics to evaluate a stock and its performance. These metrics are the following:

Return on equity shows how the firm has increased shareholders' equity and how effectively it is used. It is defined as the quotient of *Income before tax* and *Average Shareholders' Equity* as shown below. *Income before tax* is the firm's earnings before deducting tax and is calculated by subtracting all expenses from revenue, and *average shareholders' equity* is the amount that would be returned to the shareholders if the firm is liquidated.

$$\text{Return on equity} = \frac{\text{Income before tax}}{\text{Average Shareholders' Equity}}$$

Return on asset shows the firm's ability to effectively use their assets and is the quotient of *Income before tax* and *Average Total Assets*. *Average total assets* is the average of total asset over many annual periods and total assets in turn includes all the assets that the firm owns - such as cash, investments, property and plant, inventory and other equipment.

$$\text{Return on asset} = \frac{\text{Income before tax}}{\text{Average Total Assets}}$$

Gross profit margin shows the firm's revenue that is left after subtracting cost of goods sold as a percentage. It tells us how effectively a firm can execute its business operations with regards to its costs.

$$\text{Gross Profit Margin} = \frac{\text{Gross Profit}}{\text{Total Revenue}}$$

Revenue growth shows how much the firm's revenue has grown compared to last fiscal year. *Revenue* is the amount earned from selling services or goods.

$$\text{Revenue growth} = \ln \left(\frac{\text{Revenue}_t}{\text{Revenue}_{t-1}} \right)$$

All of these financial metrics are used to evaluate a firm's performance, with higher values being better than lower ones. In other words, we want higher percentages as it will separate underperforming stocks from high performing stocks.

3.4 Quality Investing

Quality investing has no universally accepted definition, however the theory as it is used in this paper will be based on Robert Novy-Marx (2014) definition:

“Quality can be viewed as alternative implementation of value-buying high quality assets without paying premium prices is just as much value investing as buying average quality asset at discount.”

A study conducted by Marx (2013) showed that by the inclusion of *gross profit-to-assets*, when predicting cross section of average returns, we can effectively separate firms with high returns from those with low returns. Firms with high gross profit are less prone to financial distress, have longer cash flow durations, and have lower operating leverage (Novy-Marx, 2013).

$$\text{Gross Profit to Asset} = \frac{\text{Gross Profit}}{\text{Total Assets}}$$

Gross profit is the profit a firm earns after subtracting the costs involved in producing and selling its products or providing its services. *Total assets* include all that the firm owns, including cash, property and plant, investments, equipment, and inventory.

3.5 The Magic Formula

The Magic Formula was created by Joel Greenblatt, a hedge fund manager at Gotham Capital, in his book *The Little Book That Beats the Market*. However, the theory will be based on his updated edition of the book: “*The Little Book That Still Beats the Market*” published the year 2010. It includes data on how the Magic Formula performed during the financial crisis of 2008 (Greenblatt, 2010).

The Magic Formula is a quantitative investment strategy based on Ben Graham and David Dodd’s theory of value investing and fundamental analysis. Joel Greenblatt’s strategy was created to help individuals to invest more easily and beat the stock market through the use of historical data. There are two key metrics in the magic formula: return on capital and earnings yield (Greenblatt, 2010, p.166, 169). The formula aims to purchase shares at a bargain price and hold them as the market over time values them at fair value as argued by the theory of value investing.

Greenblatt argues that firms with a high return on capital will have the opportunity to invest their profits at very high rates of return and that it can also yield a high rate of earnings growth. Firms that can achieve a high return on capital are likely to have some special competitive advantage. *Return on tangible capital* is calculated by dividing *earnings before interest and tax* with *tangible capital employed* (Net Working Capital + Net Fixed Assets). EBIT was used as firms are exposed to different tax rates and debt rates. Tangible capital employed was a substitute to *total assets* as the idea was to figure out how much capital is needed to execute business operations (Greenblatt, 2010, p.86, 88, 166-167).

The second metric, *earnings yield*, is calculated by dividing *earnings before interest and tax* with *enterprise value* (market value of equity + net interest-bearing debt). Greenblatt argues that this metric is better than the conventionally used *P/E ratio* (price/earnings ratio) because it includes both the price paid for an equity stake in a firm and debt used to finance business operations. Earnings yield will help to compare firms with different debt levels and tax rates (Greenblatt, 2010, p.169-170).

Greenblatt suggest that the average investor should invest in a portfolio of 20 to 30 stocks, but he also suggests that if one finds firms that have a sustainable competitive advantage, investing more in them and owning less can yield higher profits at less risk. When statistically evaluating risk, ownership of 5 stocks that you know really well and have evaluated thoroughly is worse than owning 30 stocks that you know nothing about. Greenblatt explains that this is necessarily not the case and depends on how one measures risk. If an investor has done thorough research and is certain that these 5 or 6 stocks will outperform the market, the portfolio can generate less risk and higher returns (Greenblatt, 2010, p.112-113, 141).

4 Method and Empirical Model

4.1 Investment Formula

The primary methodology that the investment formula will be based on is a fixed and random effects panel data regression model. This empirical model was chosen in order to evaluate how stock prices of different firms in the same sector changes over time depending on the value of the financial metrics mentioned in the theory section. As the metrics data is collected from annual reports, the variables have to be lagged. In this paper, the variables are based on annual reports from 2006-2022 as investors will only have access to the balance sheet of the previous year. By doing this we avoid *look-ahead bias*, which is when data that was not accessible during that time period is used in the model (Baquero, Horst and Verbeek, 2005).

The fixed effect model controls for unobserved time-invariant factors that might affect both the dependent variable and the independent variables (Studenmund, 2017). By this inclusion, we can capture firm-specific characteristics that are constant over time but may have an effect on the stock price. This is especially important due to the broad definition of the renewable energy sector in this paper as a holding company that invests in renewable energy projects differs from firms that produces energy or build and manages solar parks.

The empirical model is as follows:

$$\begin{aligned} \log(\text{Price}_{it}/\text{Price}_{it-1}) &= \beta_0 + \beta_1 \text{Gross Profit Margin}_{it-1} + \beta_2 \text{Gross Profit to Asset}_{it-1} \\ &+ \beta_3 \text{ROC}_{it-1} + \beta_4 \text{EBIT}/\text{EV}_{it-1} + \beta_5 \text{ROA}_{it-1} + \beta_6 \text{ROE}_{it-1} \\ &+ \beta_7 \log(\text{Revenue})_{it-1} + \alpha_i + \epsilon_{it} \end{aligned}$$

Where α_i is the firm-specific fixed effect and ϵ_{it} is the error term.

The random effects model assumes that the intercept of all entities is drawn from a mean around the intercepts. This model will increase our degrees of freedom but assumes that the omitted variables are not correlated with the independent variables. It also accounts for factors that fluctuate over time, such as macroeconomic conditions or policy changes.

The empirical model is as follows:

$$\begin{aligned} \log(\text{Price}_{it}/\text{Price}_{it-1}) &= \beta_0 + \beta_1 \text{Gross Profit Margin}_{it-1} + \beta_2 \text{Gross Profit to Asset}_{it-1} \\ &+ \beta_3 \text{ROC}_{it-1} + \beta_4 \text{EBIT}/\text{EV}_{it-1} + \beta_5 \text{ROA}_{it-1} + \beta_6 \text{ROE}_{it-1} \\ &+ \beta_7 \log(\text{Revenue})_{it-1} + u_i + \epsilon_{it} \end{aligned}$$

Where u_i is the random-effect and ϵ_{it} is the error term

The Hausmen tests was used as a decision-maker between these two models and is presented in the appendix section. Robust standard error was used in all regressions to mitigate heteroskedasticity.

4.2 Stock Selection

The stock selection method for the portfolio will be based on Joel Greenblatt's magic formula, but including the independent variables which were significant in the empirical model above. The dataset of stocks is sampled from Refinitiv Eikon and consists of 35 stocks. The investment period is from 2018-03-27 to 2023-03-27 and was chosen as all annual reports would have been published. Lastly, I will exclude firms with a market cap of less than 50 million USD as Joel Greenblatt suggests.

1. Include only stocks/firms in the renewable energy sector.
2. Screen for stocks using the significant variables.
3. Rank stocks high to low based on the variables above.
4. Invest in the 6 highest ranked stocks
5. Rebalance the portfolio after each annual period.

Joel Greenblatt's method involves selecting 30 stocks, however a study conducted by Vestre and Wikheim showed that purchasing a substantial amount of stocks leads to a large sum of transaction costs that in turn majorly affects overall aggregated return (Vestre and Wikheim, 2022). Due to these results, and the fact that managing a 30-stock portfolio can be challenging for the average investor, I only include 6 stocks in my portfolio.

4.3 Portfolio evaluation

To answer the second research question, the portfolio return will be compared to the top-performing clean energy Exchange-Traded fund. ETF is a pooled investment security and a SEC-registered investment company which can be bought and sold like an individual stock. They can be actively or passively managed. An ETF can be structured to track price of a commodity or a diverse and large group of securities, for example a market index of clean energy securities/stocks (*Exchange-Traded Funds (ETFs) | Investor.gov*).

ETF were specifically chosen due to readily available data, clear definition of clean energy and it being a sector-specific index. Secondly, it is also diversified, minimizing risk and is a great option for individual investors without knowledge or experience with investing in the stock

market. Assessing the investment strategy against a widely recognized benchmark will effectively explain if the strategy is superior at generating returns minimizing risk and is a great option for individual investors without knowledge or experience with investing in the stock market.

I will also compare the portfolio return with the clean energy market return and the risk-adjusted return to see how it performed at given risk and against the entire market. The risk, or beta, of the portfolio will be measured using the CAPM to calculate the risk-adjusted return.

The CAPM was used to answer the last research question, if the portfolio generates excess return, or in other words, evaluate if my portfolio's return can or cannot be explained solely by its risk/beta. An empirical regression model of CAPM was used to achieve the results:

$$R_{it} - R_{ft} = \alpha + \beta_{iM}(R_{Mt} - R_{ft}) + \varepsilon_{it}$$

Where R_{it} is the annual return of the portfolio and R_f is the average risk-free rate of US treasury bond which together represents the portfolio's excess return. β_{iM} is the risk the portfolio is exposed to compared to the market, and R_M is the annual return of the S&P Global Clean Energy Index.

5 Data

In this paper, I adopt a broad definition of the renewable energy sector, encompassing various types of firms that contribute to the production, distribution, and development of renewable energy. These firms can operate quite differently from one another, but still provide major development for technologies used to improve overall energy production and efficiency. This includes firms that involve in the generation of renewable electricity through biomass, geothermal, solar, wind and hydroelectric energy sources. Additionally, firms that provide research development, and manufacturing technologies, such as wind turbines and solar panels. Furthermore, firms specializing in financing renewable energy projects, such as venture capital firms or private equity funds are also included in our definition. Lastly, firms that construct or manage projects, such as solar farms.

Market and firm data are needed to develop the investment formula, and we want to capture how renewable energy firms performed during economic downturns and economic upswings. The years analysed in this paper were 2007-2023. These will include the economic downturn during the financial crisis of 2008 and the upswing after, and the effects of the epidemic. Greenblatt recommended a minimum of three years, and the data collected for this paper far exceeds that threshold. Price data was specifically collected 31st of December each year and is adjusted for dividends.

The selection of stocks for this study was made using Avanza's stock screener, a tool that allows one to sort through a myriad of stocks based on certain criterion. The regions screened for was Canada, USA, Sweden, Norway, France and Germany. The main objective is to identify renewable energy firms while excluding those not primarily engaged in the renewable energy production chain. Conversely, companies which mainly derive revenue from non-renewable sources were excluded. This is because the study aims to increase investments in the renewable energy sector and by excluding firms that mainly generate income through gas or oil, but also renewable energy will skew financial determinants of the stock's return.

The financial data was collected from Refinitiv Eikon and is based on renewable energy firm's earnings report which includes balance sheets, -income and cash flow statements. Historic stock data was also collected from the same source and the frequency of the data is presented in Table 5.1.

Table 5.1. Data and frequency

Data	Frequency
Log of Price	Annual
Gross Profit Margin	Annual
Gross Profit-to-asset	Annual
ROC	Annual
EBIT/EV	Annual
ROA	Annual
ROE	Annual
Log of Revenue	Annual

The variables descriptive statistics is presented in table 5.2. Gross Profit Margin, ROA, ROE and revenue growth were chosen based on the theory of value investing and its commonality traditional stock performance analysis. ROA and ROE were specifically compared by Joel Greenblatt with earnings yield and return on tangible capital, which might better explain the change in stock price more effectively than the latter. However, the main reason for the inclusion of both ROE and ROA is that the majority of free screening tools used, for example Financial Times or Yahoo Finance, include these ratios. Therefore, individuals will have an easier time screening by themselves as they do not have to calculate these metrics.

Gross Profit-to-asset was chosen based on the quantitative theory of quality investing as this variable can effectively filter out firms that doesn't have a sustainable competitive advantage. Revenue growth was also included here due to it being an effective indicator of continued growth of a firm and it suggest that the firm has some form of advantage over other competitors. Lastly, earnings yield and return on capital aligns with the magic formula and is the foundation of this paper.

Table 5.2. Descriptive statistics

Variable	Obs	Mean	Std. dev.
Log of Price	165	2.48%	1.27%
Gross Profit Margin	162	-5.97%	458.32%
Gross Profit-to-asset	162	7.28%	18.18%
ROC	167	30.25%	439.34%
EBIT/EV	164	4.74%	48.69%
ROA	165	-2.79%	24.10%
ROE	162	-6.95%	63.72%
Log of Revenue	163	12.51%	2.12%

Return on asset had an extreme outlier at 1.3 or 130%. This data-point suggest that a firm earned 30% more than their total asset; this is neither sustainable nor realistic and I therefore removed that outlier from the dataset.

The stock for the portfolio (based on 35 stocks), the market index S&P Global Clean Energy Index and the ETF SPDR S&P Kensho Clean Power ETF were all collected in Refinitiv Eikon. Lastly, the risk-free interest rate was collected from the U.S department of the treasury. All data is presented in the appendix.

6 Results and Discussion

6.1 The Investment Formula

In table 6.1.1 the initial regression results for both the random effect and the fixed effect model are presented. As stated before: robust standard errors were implemented to avoid any issues with heteroskedasticity that might introduce bias in our coefficients. The results show that return on asset (ROA), earnings yield (EBIT/EV) and return on tangible capital (ROC) are statistically significant. However, a multicollinearity test must be performed to remove any inflation of t-values as it can cause variables to become significant when they aren't. If it is not remedied, the results cannot be trusted. To detect multicollinearity, I tested for high *variance inflation factors* or VIFs. It looks at the extent to which an independent variable can be explained by all the other independent variables (Studenmund, 2017).

Table 6.1.1: Random- and fixed effect regression

Log of price	<i>Independent variable</i>	
	<i>Random effects</i>	<i>Fixed effects</i>
Growth	0.157 (0.100)	0.062 (0.104)
ROA	8.395*** (2.138)	6.017** (2.595)
ROE	-0.160 (0.421)	-0.070 (0.402)
ROC	0.160* (0.059)	-0.088** (0.024)
Gross Profit Margin	0.039 (0.132)	0.096 (0.136)
Gross Profit-to-asset	-2.093** (0.904)	0.064 (1.01)
EBIT/EV	-1.71** (0.802)	-1.01** (0.433)
Obs.	152	152
R ² Within	0.2746	0.3034
R ² Between	0.5466	0.3354
R ² Overall	0.3596	0.3182

Note: *p<0.1; **p<0.05; ***p<0.01

Return on asset (ROA) and return on equity (ROE) had the highest VIF values. Through rigorous tests with different models presented in the appendix, the final variables selected were Growth and ROA which gave the model a mean VIF value of 1.00 (table 6.1.3). ROA could be replaced with ROE, but the coefficient had less weight (i.e. smaller coefficient) and was therefore not included into the final model.

Table 6.1.2: Test for multicollinearity

Variable	VIF
ROA	5.64
ROE	3.01
EBIT/EV	2.49
Gross Profit-to-asset	2.31
Growth	2.10
Gross Profit Margin	1.33
ROC	1.14
Mean VIF	2.57

Table 6.1.3: Final Test for multicollinearity

Variable	VIF
ROA	1.00
Growth	1.00
Mean VIF	1.00

Finally, we need to perform the Hausman test to determine whether the fixed or random effect model is more appropriate for our analysis. The Hausman test evaluates whether the differences between estimators of the fixed effect model and the random effect model are statistically significant. If the test results aren't statistically significant, it implies that the random effects model is preferred (Studenmund, 2017).

The Hausman test yielded a chi-2 value of 0.43 and a p-value (prob > chi2) of 0.8047 (Appendix 2). Since the p-value is greater than 0.05, we fail to reject the null hypothesis and proceed with the random effect regression model. This model will increase the degrees of freedom and account for variations over time, such as changes in macroeconomic conditions or policy shifts, providing more efficient and consistent estimators.

That leads us to the final regression model in which the correlation between these two remaining variables to changes in stock price is investigated, presented in table 6.1.4. Return on asset (ROA) is shown to be statistically significant at less than a 1% level and revenue growth (Growth) at less than a 10% level. The result shows that these variables can effectively explain the increase in stock price over time. Even though revenue growth was only significant at less than 10%, theory supports the notion that it has a positive effect on changes in stock price.

Table 6.1.4: Final model: Random effect regression

<i>Dependent variable: Log of price</i>	<i>Independent variable</i>
Growth	0.106* (0.061)
ROA	4.931*** (0.982)
Obs.	157
R ² Within	0.3067
R ² Between	0.3086
R ² Overall	0.3136
Note:	*p<0.1; **p<0.05; ***p<0.01

Earnings yield (EBIT/EV) was removed from the model even if it was statistically significant because of its high correlation with return on asset (ROA). Also, it had a negative coefficient, implying that a renewable energy firm earning less in comparison to its enterprise value is more “valuable” than a firm that earns more compared to its enterprise value. At first glance, this might not seem theoretically sound. However, there is possibly an empirical reason that can explain this result. A firm with lower EBIT to enterprise value might have larger debt-ratios paying more in amortization with prospects of future growth. Due to this ambiguity, earnings yield was not included in the Investment Formula.

Return on tangible capital (ROC) was removed as it was only statistically significant due to multicollinearity which inflated its t-value.

The coefficient of return on asset (ROA) has the highest weight and therefore, we can conclude that it is the most influential variable. This is theoretically and empirically sound as a higher return on a firm's assets indicates efficient use of their assets. The renewable energy sector is

characterized by its reliance on assets, such as solar parks or material-producing factories, which in turn explains the large weight of the coefficient.

Based on these findings, the final investment formula is

$$\text{Return on asset} = \frac{\text{Income before tax}}{\text{Average Total Assets}}$$

$$\text{Revenue growth} = \ln(\text{Revenue}_t / \text{Revenue}_{t-1})$$

where ROA takes higher priority due to its significant weight, while revenue growth can be used to distinguish between firms with identical ROA values. If average total asset is not publicly available then the total assets of that year, which is presented in the balance sheet, should be used instead.

6.2 Portfolio performance

The investment formula outperformed both the top-performing ETF and the overall market index for renewable energy firms as presented in table 6.3. The ETF's return is adjusted for management costs of 0,45% per year.

Table 6.2. Portfolio 2018-2023

	Average annual return
Portfolio	25.47%
SPDR S&P Kensho Clean Power ETF	13.63%
S&P Global Clean Energy Index	14.91%

As presented in figure 6.1 the investment formula did not outperform the market index in the first period, however it did outperform it in the four following periods. SPDR Kensho Clean Power ETF underperformed against the portfolio and market in all periods except during the third period. These results suggest that investing according to my formula was a better strategy compared to investing in the ETF.

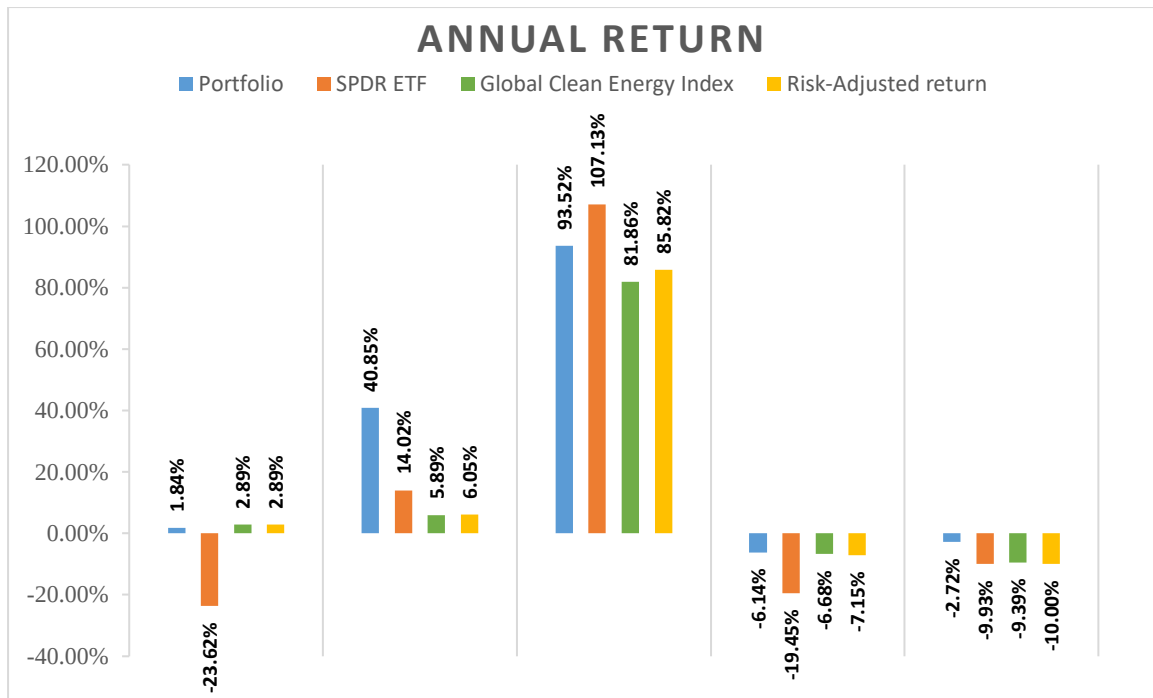


Figure 6.1. Annual Return 2018-2023

If we compare the market index and my formula, the overall risk taken were almost equal in downturns with the market only performing better in the first period. Given the small differences and the large return difference, it suggests that my formula is the overall better investment strategy. The risk-adjusted return, given a beta of 1,05 presented in section 6.3, only outperformed my portfolio in the first period. These results imply that my formula has some degree of risk-adjusted outperformance compared to the market but only in four out of the five periods.

6.3 Excess Return and Considerations

In table 6.2 below, the results from the CAPM regression are presented. The average beta of the portfolio was 1.05 which implies that it is exposed to more risk than the overall market and Jensen's alpha is not significant. Therefore, the portfolio does not generate any excess return in comparison to the risk. These results are in line with the efficient market hypothesis as the larger return is explained by the increased risk it is exposed to.

Table 6.3. CAPM regression results.

<i>Independent variable: Excess return</i>	<i>Independent variable</i>
Beta	1.05** (0.22)
Alpha	0.099 (0.083)
R ²	0.8849
Note:	*p<0.1; **p<0.05; ***p<0.01

So, taken together, do these findings support the use of this formula as a sound investment strategy? The formula does not generate excess return, but if the number of periods in my portfolio had been increased, I do think it would have generated excess return as it did outperform the risk-adjusted return each period except for the first one. Furthermore, the formula outperformed the in terms of return highest ranked EFT which in general is readily available for individuals to invest in. My formula might not be an “easier” method, but it is by no means difficult to apply. All one has to do is to use a free screener tool, provided by Financial Times or Yahoo Finance, and select ROA as a criterion. Then you limit market cap to \$50 million or greater and filter from highest to lowest ROA. Lastly, my portfolio outperformed the market and the ETF in average annual return by a large margin over time. In realized returns, my portfolio was the best.

Besides creating the formula, another aim of this paper was for it to be easily comprehensible to the average investor. This is not easily measured as there aren’t any empirical data or studies that test how Joel Greenblatt’s formula performed for the average investor and the difficulties that come with it. However, as my method of selecting stocks is the same as Joel Greenblatt’s, with some minor alterations, it should be at least as easy. My investment formula includes variables that are found in the majority of screener tools, which makes it even more efficient than Joel Greenblatt’s method of selecting stocks.

7 Conclusion

The purpose of this paper was to create an investment formula based on Joel Greenblatt's Magic Formula in order to aid individuals in their selection of stocks in the renewable energy sector. Such a formula would have the potential to be used as a tool and encourage investments in a sector that is pivotal for a sustainable society. Based on panel data regression and conventional finance theory, the key metrics were determined to be return on asset and revenue growth, with an emphasis on the former of the two. With an average return of 25.46%, it outperformed the top-performing clean energy ETF and the global market index for renewable energy firms. Despite not generating any excess return, it did beat the risk-adjusted return in four out of five periods. To increase the statistical robustness, future researchers should include a larger subset of stocks in the selection process and extend the time period to include 4-5 more years of performance when creating the portfolio. I believe it would prove that my formula does indeed generate excess return as four of the five periods did outperform both the market and the risk-adjusted return. In conclusion, my formula is a high-yielding investment strategy for investing in the renewable energy sector.

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9 Appendix

A1 Stocks included in the Panel Data

Table A1: Stocks 2007-2023

2007-2023	
Ellomay Capital LTD (XNYS: ELLO)	Voltalia SA (XPAR: VLTSA)
PNE AG (XFRA: PNE3)	Canadian Solar INC (XNAS: CSIQ)
Boralex INC (XTSE: BLX)	Encavis AG (XFRA: ECV)
Nordex SE (XETR: NDX1)	Enphase Energy INC (XNAS: FSLR)
First Solar INC (XNAS: ENPH)	REC Silicon ASA (XOSL: RECSI)

A2 Statistics of the Panel Data

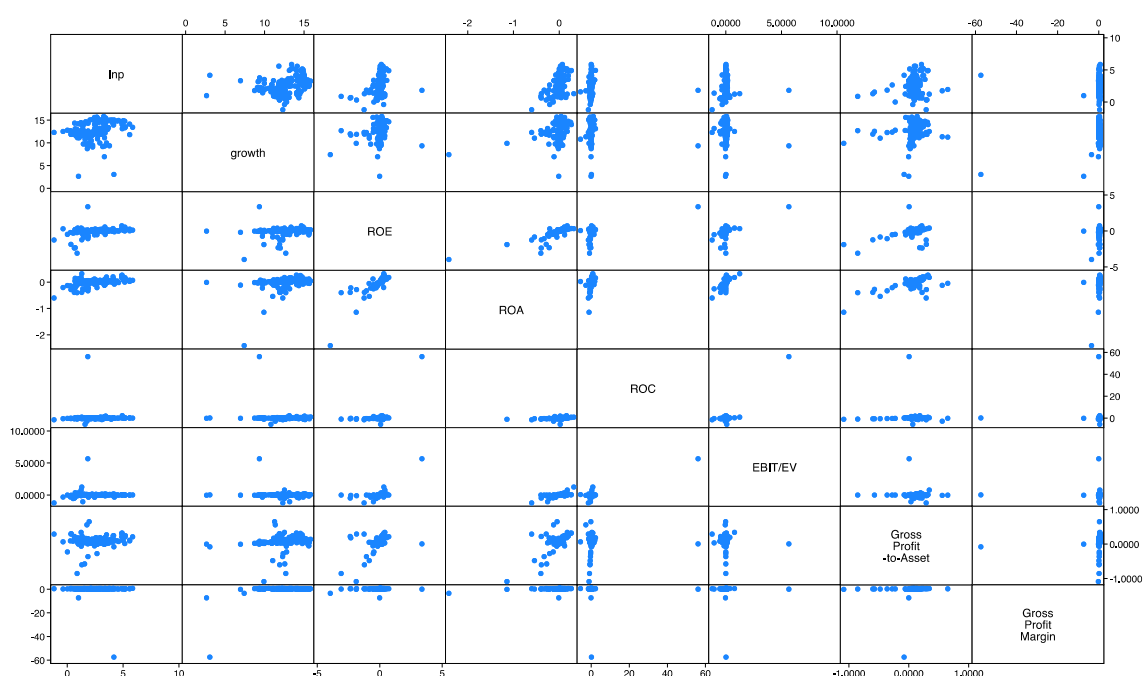


Figure A2: Scatter plot matrix.

Table A2.1 initial regression results from random effect and fixed effect

Log of price	Independent variable	
	Random effects	Fixed effects
Growth	0.157 (0.100)	0.062 (0.104)
ROA	8.395*** (2.138)	6.017** (2.595)
ROE	-0.160 (0.421)	-0.070 (0.402)
ROC	0.160* (0.059)	-0.088** (0.024)
Gross Profit Margin	0.039 (0.132)	0.096 (0.136)
Gross Profit-to-asset	-2.093** (0.904)	0.064 (1.01)
EBIT/EV	-1.71** (0.802)	-1.01** (0.433)
Obs.	152	152
R ² Within	0.2746	0.3034
R ² Between	0.5466	0.3354
R ² Overall	0.3596	0.3182
Note:	*p<0.1; **p<0.05; ***p<0.01	

Table A2.2 VIF values of the independent variables

Variable	VIF
EBIT/EV	15.18
ROA	9.43
ROC	8.21
ROE	3.97
Growth	2.10
Gross Profit-to-asset	2.30
Gross Profit Margin	1.33
Mean VIF	6.07

Table A2.3 VIF values after removing high VIF variables.

Variable	VIF
ROA	2.48
EBIT/EV	2.14
Gross Profit-to-asset	1.87
Growth	1.56
Mean VIF	2.02

Table A2.4 Final models VIF value.

Variable	VIF
ROA	1.00
Growth	1.00
Mean VIF	1.00

Table A2.4 Test 1 for the final model

Log of price	Independent variable	
	Random effects	Fixed effects
Growth	0.107* (0.846)	0.089 (0.070)
ROA	6.202*** (0.846)	6.080** (0.897)
EBIT/EV	-1.038** (0.480)	-1.014** (0.435)
Gross Profit-to-asset	-0.316 (0.487)	-0.111 (0.561)
Obs.	156	156
R ² Within	0.3099	0.3106
R ² Between	0.3556	0.3254
R ² Overall	0.3315	0.3224

Note: *p<0.1; **p<0.05; ***p<0.01

Table A2.5 Test 2 for the final model

Log of price	Independent variable	
	Random effects	Fixed effects
Growth	0.100 (0.063)	0.088 (0.070)
ROA	5.971*** (1.032)	5.998*** (1.129)
EBIT/EV	-0.973** (0.472)	-0.990** (0.473)
Obs.	156	156
R ² Within	0.3103	0.3105
R ² Between	0.3202	0.3127
R ² Overall	0.3219	0.3189
Note:	*p<0.1; **p<0.05; ***p<0.01	

Table A2.6 Test 3 for the final model

Log of price	Independent variable	
	Random effects	Fixed effects
Growth	0.106* (0.061)	0.091 (0.069)
ROA	4.931*** (0.982)	4.967*** (1.013)
Obs.	157	157
R ² Within	0.3067	0.3070
R ² Between	0.3086	0.2972
R ² Overall	0.3136	0.3065
Note:	*p<0.1; **p<0.05; ***p<0.01	

Table A2.7 Results from Hausmen test.

	(b) Fixed	(B) Random	(b-B) Difference	Sqrt(diag(V_b-V_B)) Std. err.
Growth	0.092	0.106	-0.014	0.023
ROA	4.967	4.931	0.036	0.144
Chi2(3)	0.43			
Prob > chi2	0.8047			

Table A2.8 Final model: Random effects.

<i>Dependent variable: Log of price</i>	<i>Independent variable</i>
Growth	0.106* (0.061)
ROA	4.931*** (0.982)
Obs.	157
R ² Within	0.3067
R ² Between	0.3086
R ² Overall	0.3136
Note:	*p<0.1; p**<0.05; ***p<0.01

A3 Portfolio

Table A3.1 35 stocks for selection process

BEAM GLOBAL (XNAS:BEEM)	MAXEON SOLAR TECHNOLOGIES, LTD. (XNAS:MAXN)
Arise AB (XSTO:ARISE)	Eolus Vind AB (publ) (XSTO:EOLU B)
POLARIS RENEWABLE ENERGY INC. (XTSE:PIF)	ELLOMAY CAPITAL LTD (XNYS:ELLO)
ALTIUS RENEWABLE ROYALTIES CORP. (XTSE:ARR)	VAST RENEWABLES LIMITED (XNAS:VSTE)
Energiekontor AG (XFRA:EKT)	Voltalia SA (XPAR:VLTSA)
PNE AG (XFRA:PNE3)	OX2 AB (publ) (XSTO:OX2)
CANADIAN SOLAR INC. (XNAS:CSIQ)	INNERGEX RENEWABLE ENERGY INC. (XTSE:INE)
RENEW ENERGY GLOBAL PLC (XNAS:RNW)	ENLIGHT RENEWABLE ENERGY LTD (XNAS:ENLT)
SMA Solar Technology AG (XETR:S92)	Fluence Energy Inc (XNAS:FLNC)
ARRAY TECHNOLOGIES, INC. (XNAS:ARRY)	BORALEX INC. (XTSE:BLX)
SUNRUN INC. (XNAS:RUN)	Encavis AG (XFRA:ECV)
Nordex SE (XETR:NDX1)	SOLAREEDGE TECHNOLOGIES, INC. (XNAS:SEDG)
BROOKFIELD RENEWABLE CORPORATION (XNYS:BEPC)	Neoen SA (XPAR:NEOEN)
ENPHASE ENERGY, INC. (XNAS:ENPH)	Hydro One Limited (XTSE:H)
FIRST SOLAR, INC. (XNAS:FSLR)	Spruce Power Holding Corporation (XNYS:SPRU)
Magnora ASA (XOSL:MGN)	Edda Wind ASA (XOSL:EWIND)
Cloudberry Clean Energy ASA (XOSL:CLOUD)	REC Silicon ASA (XOSL:RECSI)
Scatec ASA (XOSL:SCATC)	

Table A3.2 data on portfolio

	Annual return	Weights	
Portfolio 1 2018-03-27 - 2019-03-26)			
Eolus Vind AB (publ) (XSTO:EOLU B)	37.96%	16.65%	
Energiekontor AG (XFRA:EKT)	14.84%	16.67%	
PNE AG (XFRA:PNE3)	-25.11%	16.67%	
SOLAREEDGE TECHNOLOGIES, INC. (XNAS:SEDG)	-43.28%	16.67%	
FIRST SOLAR, INC. (XNAS:FSLR)	-37.56%	16.67%	
Scatec ASA (XOSL:SCATC)	64.24%	16.67%	1.84%
Portfolio 2 (2019-03-27 - 2020-03-26)			
Eolus Vind AB (publ) (XSTO:EOLU B)	71.40%	16.67%	
Arise AB (XSTO:ARISE)	41.74%	16.65%	
CANADIAN SOLAR INC. (XNAS:CSIQ)	-4.36%	16.67%	
SOLAREEDGE TECHNOLOGIES, INC. (XNAS:SEDG)	89.90%	16.67%	
Hydro One Limited (XTSE:H)	20.35%	16.67%	
Magnora ASA (XOSL:MGN)	26.07%	16.67%	40.85%
Portfolio 3 (2020-03-27 - 2021-03-26)			
Eolus Vind AB (publ) (XSTO:EOLU B)	70.60%	16.67%	
CANADIAN SOLAR INC. (XNAS:CSIQ)	98.91%	16.67%	
SOLAREEDGE TECHNOLOGIES, INC. (XNAS:SEDG)	111.47%	16.65%	
ENPHASE ENERGY, INC. (XNAS:ENPH)	143.72%	16.67%	
Magnora ASA (XOSL:MGN)	120.33%	16.67%	
Hydro One Limited (XTSE:H)	16.13%	16.67%	93.52%
Portfolio 4 (2021-03-29 - 2022-03-25)			
Energiekontor AG (XFRA:EKT)	36.35%	16.67%	
ARRAY TECHNOLOGIES, INC. (XNAS:ARRY)	-77.89%	16.67%	
SOLAREEDGE TECHNOLOGIES, INC. (XNAS:SEDG)	19.54%	16.67%	
ENPHASE ENERGY, INC. (XNAS:ENPH)	19.54%	16.65%	
FIRST SOLAR, INC. (XNAS:FSLR)	-7.31%	16.67%	
Magnora ASA (XOSL:MGN)	-27.04%	16.67%	-6.14%
Portfolio 5 (2022-03-28 - 2023-03-27)			
POLARIS RENEWABLE ENERGY INC. (XTSE:PIF)	1.95%	16.67%	
Energiekontor AG (XFRA:EKT)	-27.90%	16.67%	
CANADIAN SOLAR INC. (XNAS:CSIQ)	7.84%	16.67%	
ENPHASE ENERGY, INC. (XNAS:ENPH)	1.04%	16.67%	
FIRST SOLAR, INC. (XNAS:FSLR)	97.40%	16.67%	
Spruce Power Holding Corporation (XNYS:SPRU)	-96.76%	16.65%	-2.72%

Table A3.3 ETF: SPDR S&P Kensho Clean Power ETF

Year	SPDR S&P Kensho Clean Power ETF	Fee
2018-2019	-23.62%	0.45%
2019-2020	14.02%	0.45%
2020-2021	107.13%	0.45%
2021-2022	-19.45%	0.45%
2022-2023	-9.93%	0.45%
Average	13.63%	

Table A3.4 Market index: S&P Global Clean Energy Index

Year	S&P Global Clean Energy Index Return
2018-2019	2.89%
2019-2020	5.89%
2020-2021	81.86%
2021-2022	-6.68%
2022-2023	-9.39%
Average	14.91%